

PtrTrans: CodeWeaver evidence profile

This profile audits every empirical surface identified for PtrTrans and records the maximum scientifically defensible CodeWeaver comparison. Status: public_artifact_license_restricted. Crown-16 is disjoint from CRUST-Bench and cannot be redistributed without a license.

Evidence boundary

Paper values and CodeWeaver measurements remain separate unless subject identity, revision, denominator, and fixed oracle are verified. Missing values are not converted to zeros.

Existing result

No separate compatible CodeWeaver result package.

Complete empirical-surface audit

key	surface	denominator	metrics	status
ptrtrans	Tables 1-2	16 Crown projects	subject size and fuzz-test coverage	exact_reference_public_read_only
ptrtrans	Tables 3-4	16 Crown projects	lints, unsafe usage, compiled and equivalent functions	exact_reference_public_read_only
ptrtrans	Table 5	10 small Crown projects x 5 ablations	compiled and equivalent functions	exact_reference_public_read_only

Inclusion and execution decision

paper_id	source_url	empirical_scope	crust_role	codeweaver_status	reason	existing_result
arXiv:2510.10956	https://arxiv.org/abs/2510.10956	16 Crown projects	citation only	public_artifact_license_restricted	Crown-16 is disjoint from CRUST-Bench and cannot be redistributed without a license	

ptrtrans_reference

surface	scope	system	metric	value
Table 3	Crown-16 total	Crown	lint alerts	6802
Table 3	Crown-16 total	PR2	lint alerts	4135
Table 3	Crown-16 total	PtrTrans	lint alerts	349
Table 3	Crown-16 total	Crown	unsafe usages	141866
Table 3	Crown-16 total	PR2	unsafe usages	134185

surface	scope	system	metric	value
Table 3	Crown-16 total	PtrTrans	unsafe usages	85
Table 4	small projects	FLOURINE	compiled/equivalent percent	69.9/52.3
Table 4	small projects	PtrTrans	compiled/equivalent percent	98.3/81.6
Table 4	large projects	FLOURINE	compiled/equivalent percent	64.0/14.2
Table 4	large projects	PtrTrans	compiled/equivalent percent	85.9/67.9
Table 5	small-10 average	PtrTrans_PS	compiled/equivalent percent	89.3/59.5
Table 5	small-10 average	PtrTrans_PU	compiled/equivalent percent	84.6/52.9
Table 5	small-10 average	PtrTrans_RA	compiled/equivalent percent	87.9/61.9
Table 5	small-10 average	PtrTrans_EC	compiled/equivalent percent	66.0/50.8
Table 5	small-10 average	PtrTrans	compiled/equivalent percent	100/81.6